

AdvocateAuroraHealth

Title: AAH HIT Data Backup and Recovery		Document Number: 2488
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I. PURPOSE

To protect Advocate Aurora Health (AAH) data from loss or corruption due to a disaster or other business interrupting event.

II. SCOPE

This policy applies to all Advocate Aurora Health, Inc. AAH team members and third parties who interact with AAH data residing on systems, networks, and applications hosted in AAH Data Centers and with providers of Cloud Computing Services IaaS and PaaS. SaaS is dependent on contract requirements. This policy is for data backup and recovery purposes only and does not apply to long-term compliance or legal hold retention requirements.

III. DEFINITIONS/ABBREVIATIONS

AAH Data: All data that is owned by AAH or owned by third parties but provided to AAH. For clarification purposes only, data remains the property of AAH even if it is in the possession and under the control of a Cloud Computing Services provider or a third party (e.g., a third-party service provider and/or its subcontractors).

AAH Data Center: Hardened facility owned and operated by AAH or contracted through a third-party provider. AAH data centers are used for the remote storage, processing, and distribution of AAH data.

Cloud Computing: the practice of using a network of remote servers hosted on the internet to store, manage, and process data, rather than a local server or a personal computer.

Cloud Computing Services Platform Types

- **Infrastructure as a Service (IaaS):** Allows for ability to configure and manage the hardware for your application and is most flexible cloud service.
- **Platform as a Service (PaaS):** The platform management is handled by the cloud provider and the focus is on the application development.
- **Software as a Service (SaaS):** Users pay for the software as they use it on a subscription model; a pay-as-you-go pricing model.

Third-Party Hosted: Applications and data which AAH has contracted for and hosted off-premises by a third- party vendor.

Data Backup: The copying of data to media that can be stored both on and/or offsite and be used to restore corrupted or lost data or to recover entire systems, files and databases in the event of a disaster or business continuity event.

Air Gap Backup: An air gap is a technical configuration of the backup environment where backup data is stored offline and separate from the production environment.

Data Backup Encryption: For security purposes it is the process of encoding backup data in such a way that only authorized parties can read it.

Data Backup Retention: The amount of time a data backup will be retained until destroyed or over-written.

Data Recovery: The restoration of data from backup media to restore programs and production data to the state that existed at the time of the last safe backup.

Data Replication: The process of copying data from one location to another. Provides near real-time copies of data in the event of a data impacting disaster or business continuity event.

Disaster Recovery Tiers: (described in the HIT Disaster Recovery Tiers Standard): Tiers which define the time HIT must recover an application or system before business is being negatively impacted and downtime procedures are no longer practical.

Geographically Diverse Facilities: facilities which are used to store data backups and/or replicated data must be far enough apart that they are not subject to most of the same physical risks – e.g. storms, earthquakes, power outages, etc.

Hardened Facility: facility with strictly controlled environment including physical security, climate control, redundant power, redundant cooling, fire detection and

extinguishing systems. These facilities are designed to withstand natural and man-made disasters.

Removable Media: Removable media devices include, but are not limited to the following:

1. Optical Disks such as Blu-ray, DVDs, CD-ROMs
2. Memory Cards including but not limited to Compact Flash card, Mobile Phone SIM Cards, SD Card, Memory Stick
3. Portable USB flash drives (also known as pen drives, thumb drives, jump drives or key drives).
4. External Hard Drives – DE, EIDE, SCSI, SSD
5. USB card readers that allow connectivity to a PC.
6. Portable MP3 and MPEG-playing music and media player-type devices.
7. Any hardware that provides connectivity to USB devices through means such as wireless (Wi-Fi, WiMAX, IrDA, Bluetooth, among others) or wired network access.
8. Other external devices which contain removable media capabilities

IV. **POLICY**

Backups of data and applications residing on systems and networks hosted in AAH Data Centers, third-party hosted, and by providers of Cloud Computing services (IaaS and PaaS), must be executed and retained to meet application recoverability requirements per the HIT Data Backup Retention Standard and HIT Disaster Recovery Tiers Standard.

Note: Providers of cloud computing services (SaaS) that partner with AAH for solutions per BAA or contract must follow industry standards and best practices for data backup, retention, encryption, and storage of backups.

A. Data Backup and Replication

1. Data backups must be created per the defined workload and associated backup schedule in the HIT Data Backup Retention Standard.
2. Data Replication to a separate geographically diverse and hardened facility must be used when necessary to meet application

recoverability requirements. Refer to HIT Disaster Recovery Tiers Standard for specific recovery time objectives.

B. Data Backup Retention

1. Data backups must be retained per the defined workload and associated backup schedule in the HIT Data Backup Retention Standard.
2. Data Backups must be retained to meet the business application's recoverability time requirements per the assigned HIT Disaster Recovery Tier.

C. Data Backup Encryption

1. Data Backups at rest must be encrypted.

D. Data Backup Location and Storage

1. Data backups must reside in separate geographically diverse and hardened facilities

E. Data Backup Quality Verification

1. A data backup execution report must be documented to confirm successful completion and failures must be remediated
2. Data restoration must be performed to validate integrity and content of backups. Refer to HIT Data Backup Retention Standard for specific quality verifications.
3. Database integrity checks must be performed to validate quality. Refer to HIT Data Backup Retention Standard for specific quality verification requirements.

F. Data Backup and Recovery Systems

1. A redundant backup system in a separate hardened facility must be maintained and can be activated without loss of information or disruption to operations.
2. Data backup restores must be sample tested when any major upgrade is made to the backup system.

G. Data Backup and Recovery Procedures and Technology Documentation must be maintained by the HIT Storage Area Network team, specifically:

1. Data backup and recovery procedures must be documented
2. Storage and media, backup technology and locations must be documented.
3. Third-party backup storage locations and processes to access stored data must be documented.
4. Periodic reviews of backup and recovery processes and procedures must be performed.

5. Documentation must reside in two geographically diverse facilities.

V. PROCEDURES

N/A

VI. CROSS REFERENCES

- A. AAH HIT Disaster Recovery Planning Policy
- B. HIT Data Backup Retention Standard
- C. HIT Disaster Recovery Tiers Standard

VII. RESOURCES AND REFERENCES

Epic Business Continuity Technical Solutions Guide

NFPA 1600 Standard on Disaster/Emergency Management and Business Continuity/Continuity of Operations Program

NIST 800-53r5 CP-2, CP-4, CP-9, CP-10

HIPPA CFR 164.308

PCI DSS 3.2.1 3.4, 9.5.1, 12.10.1

VIII. ATTACHMENTS

N/A